

VerifyFlow

AI-Powered Document Verification

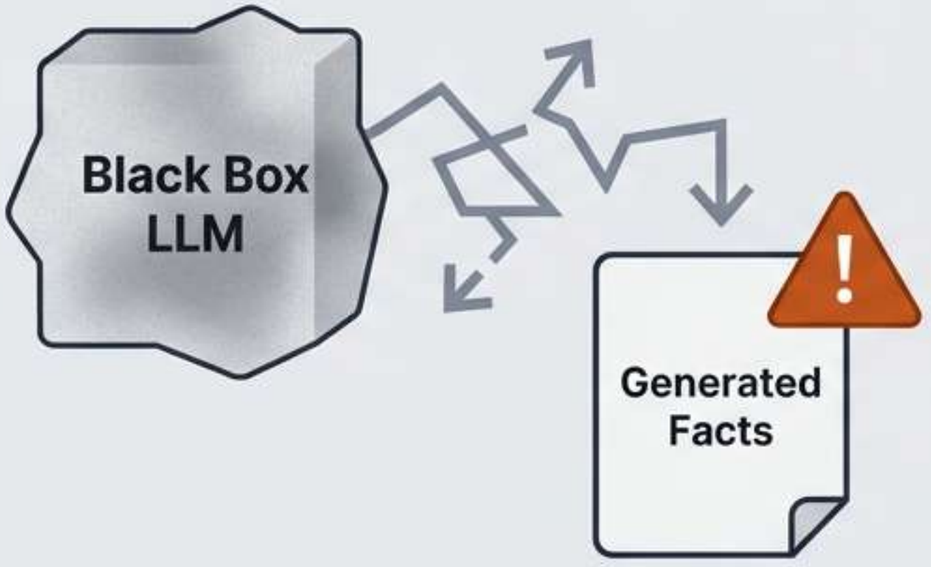
Postulate

The model can think. The evidence gets the final say.

Nebius x NVIDIA Hackathon

Gemini Notebook

Trust Must Be Deterministic, Not Generated



Enterprise LLMs **hallucinate** facts in complex legal and financial documents.

Raw generative text is a **liability**.



Reasoning Engine



Decision Engine

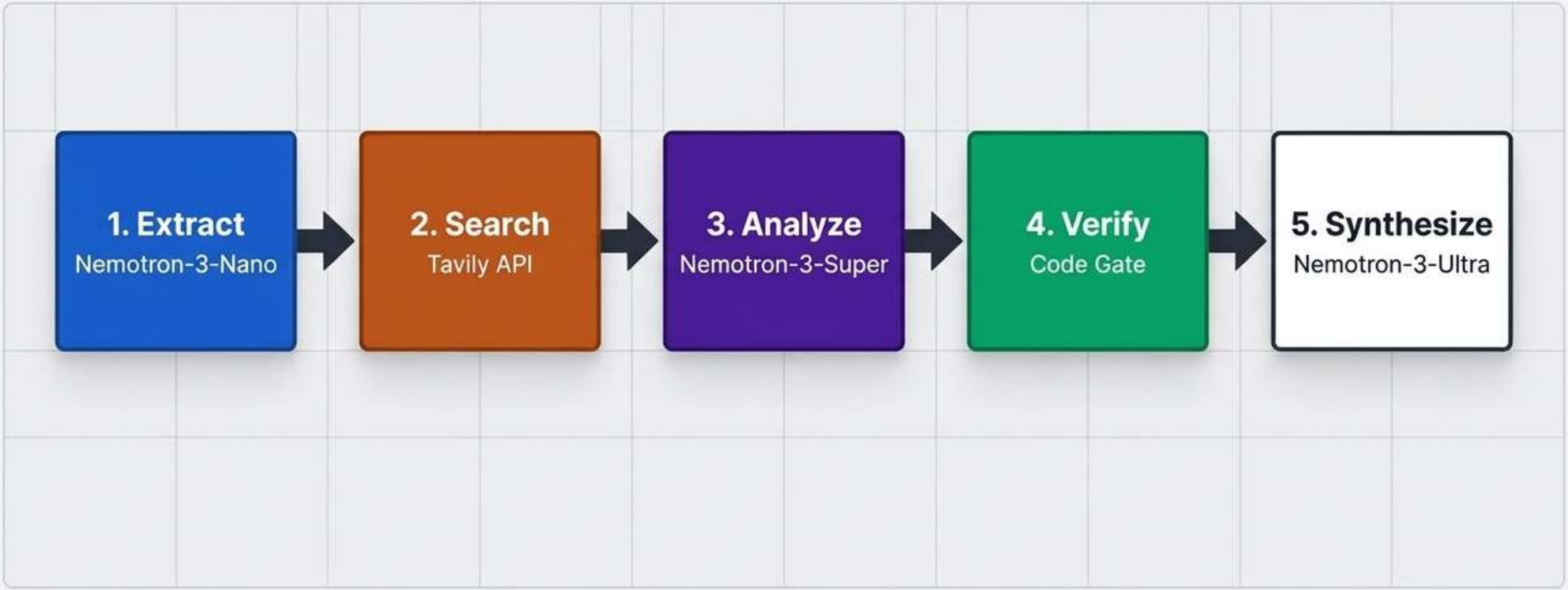
Separate reasoning from decision-making.

Use LLMs to **extract and analyze**, but rely strictly on **deterministic code** to issue the **final verification verdict**.

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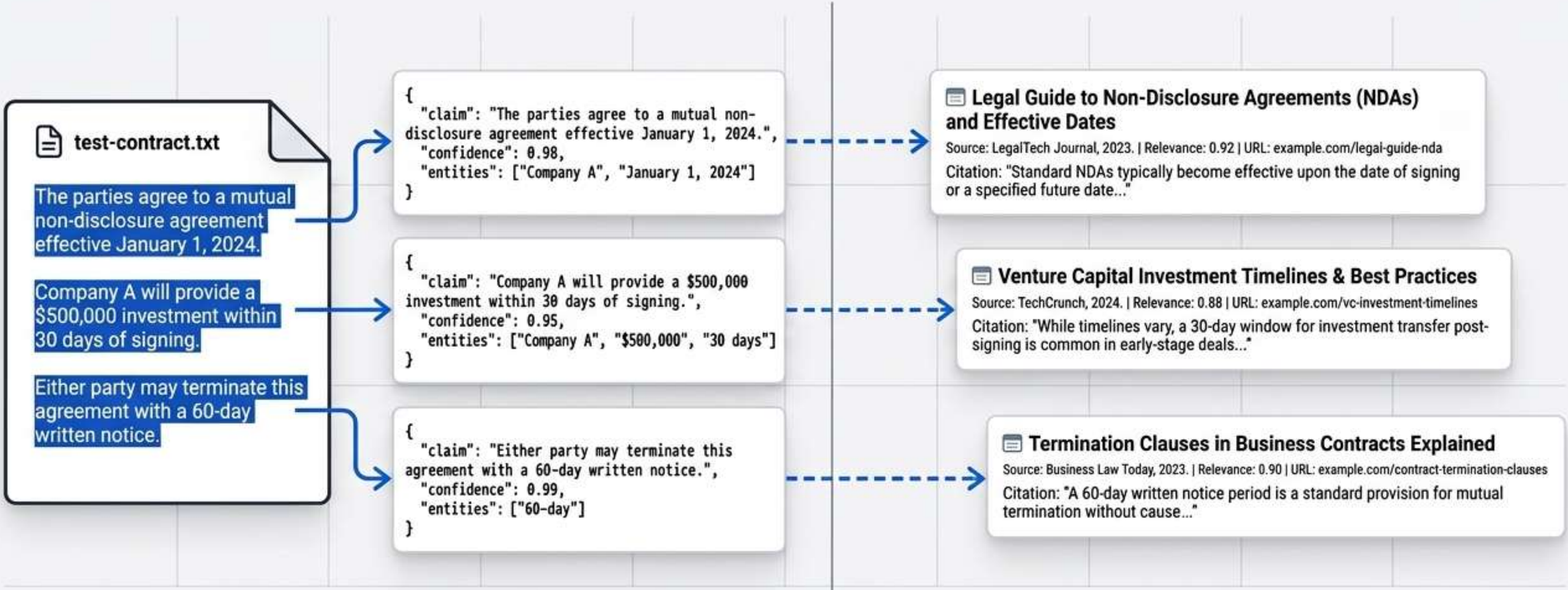
The 5-Stage Verification Architecture

A modular pipeline designed to cross-reference extracted claims against live web evidence while maintaining strict cost controls.



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Stages 1 & 2: High-Volume Extraction & Grounding

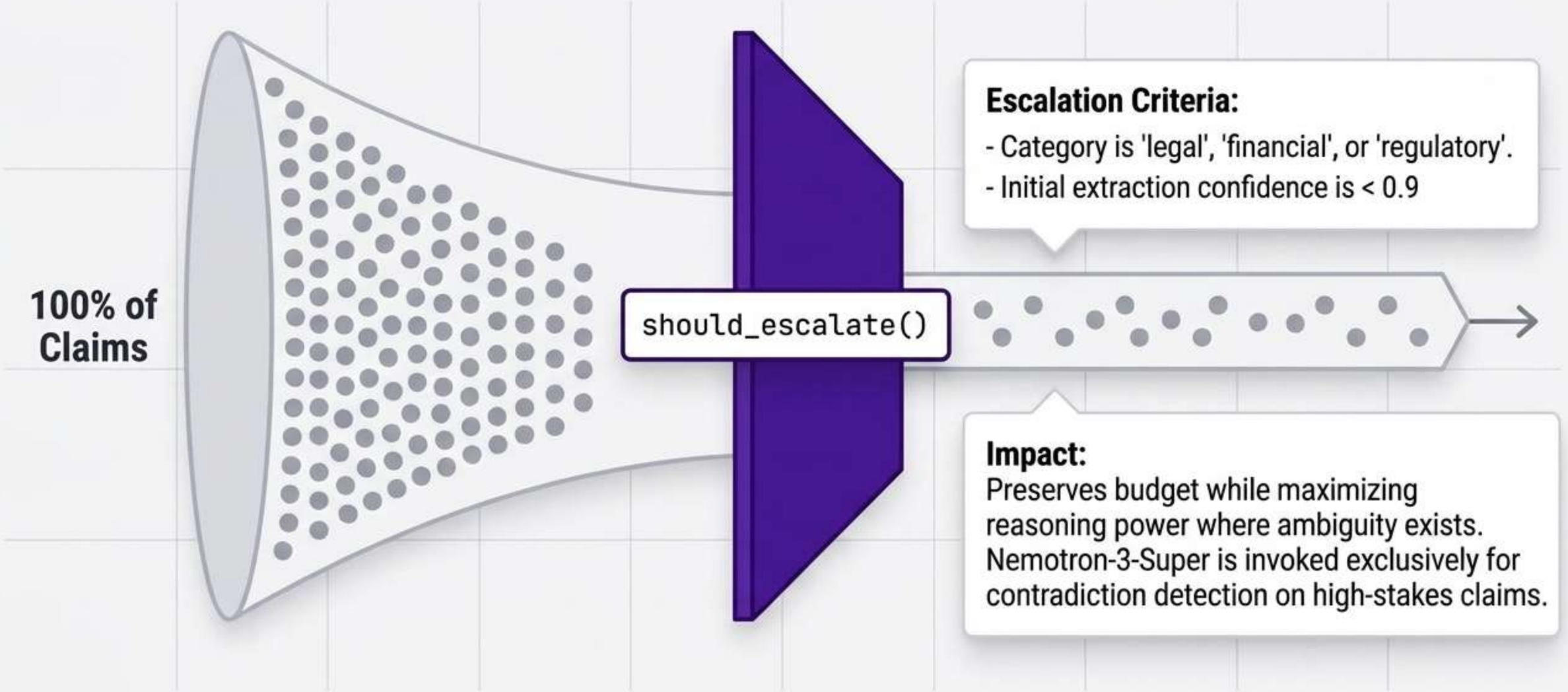


Nemotron-3-Nano: Parses documents up to 500,000 characters to extract structured claims, confidence scores (0.0 - 1.0), and named entities. Handled by the fastest, cheapest tier.

Tavily Search API: Automatically generates advanced search queries for every extracted claim, pulling real-time URLs, citations, and relevance scores from the live web.

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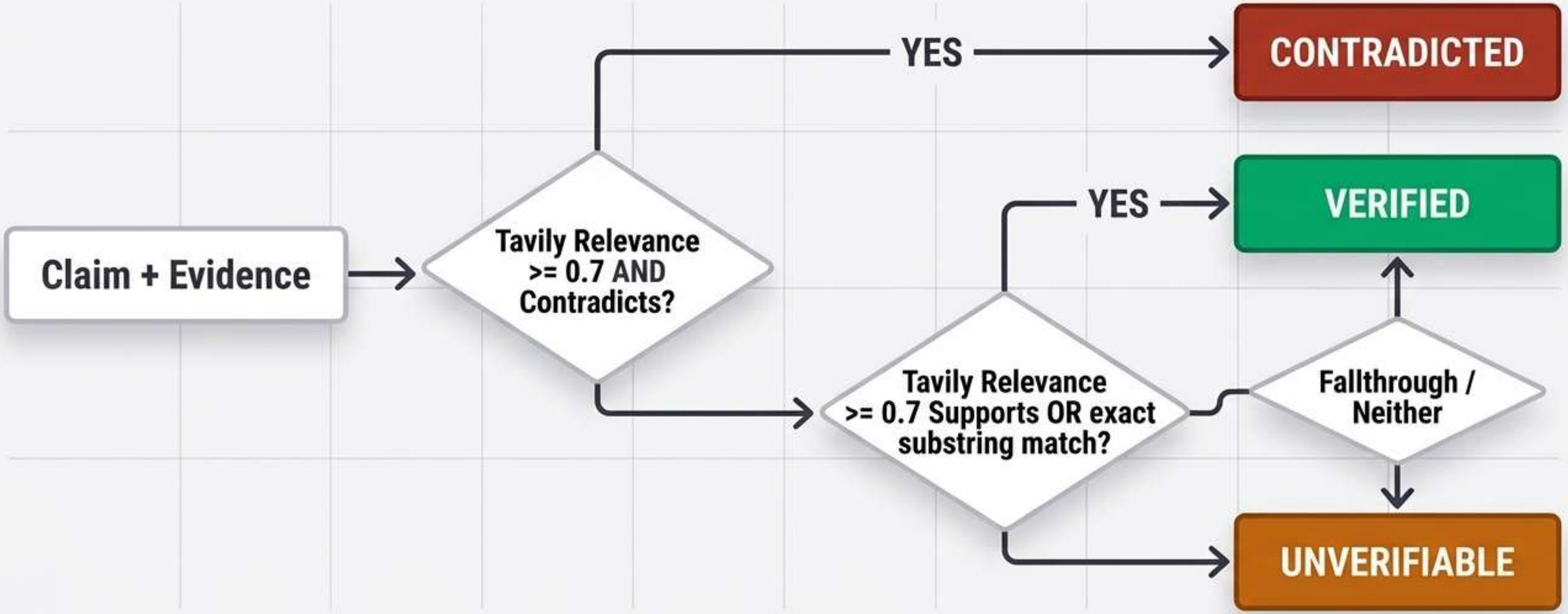
Stage 3: Deep Analysis Escalation



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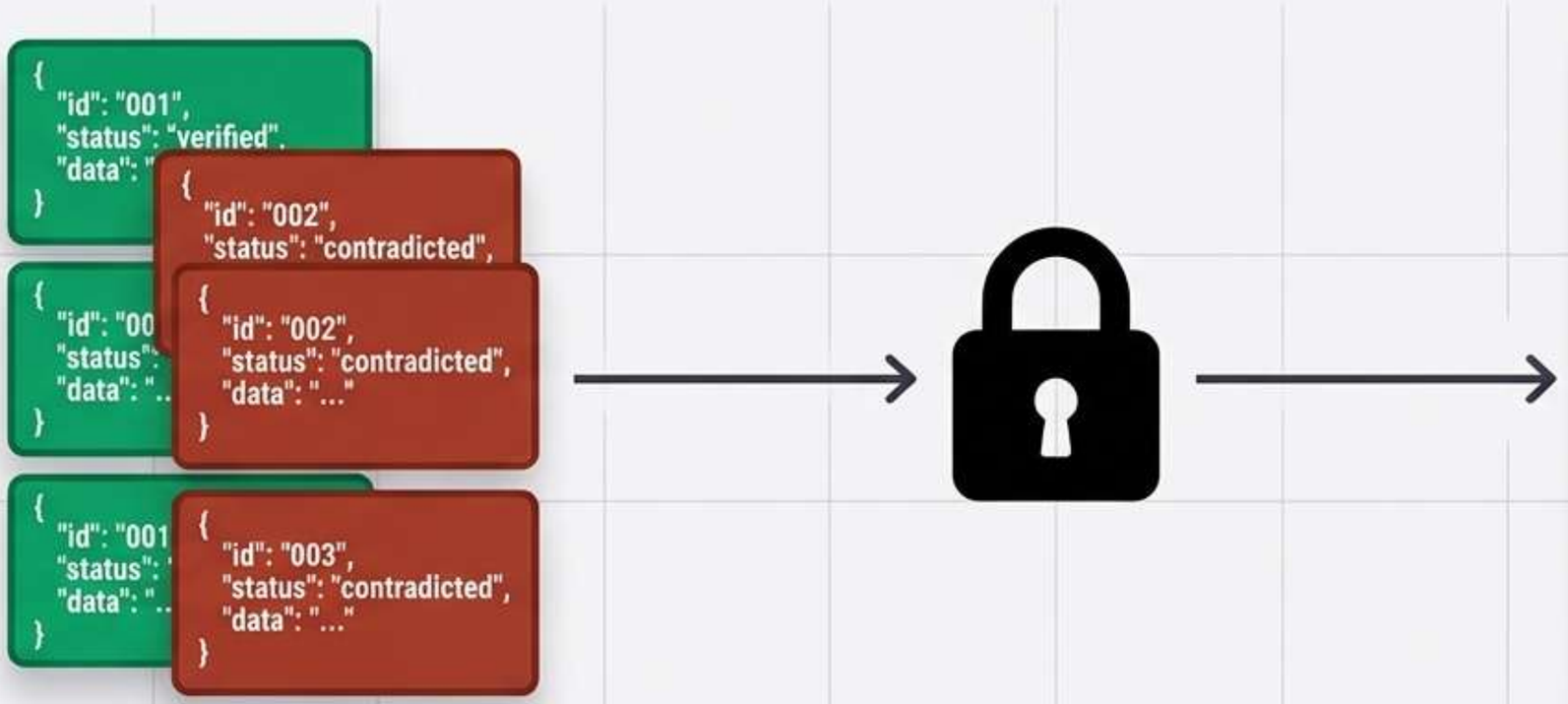
Stage 4: The Deterministic Trust Gate

ZERO hallucination risk at the decision layer. LLMs are entirely removed from this stage. Hard-coded thresholds dictate the final state based solely on retrieved evidence.



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Stage 5: Final Synthesis



The Rule of One:
Nemotron-3-Ultra (550B parameters) is called exactly once at the very end of the pipeline.

The Output:
It reads the deterministic verdicts and generates a professional, human-readable executive summary.

The Constraint:
Prompted with a strict constraint: “Do NOT change any verification status — those decisions are final.”

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Unit Economics: Nebius Tiered Routing

Operating a complex multi-agent pipeline at scale while maintaining a $50 hackathon budget.

Nano (30B)

Cost:
$0.06 / 1M Input Tokens

Volume:
85% of Pipeline Traffic

Role:
High-speed claim extraction & classification.

Super (120B)

Cost:
$0.30 / 1M Input Tokens

Volume:
10% of Pipeline Traffic

Role:
Deep contradiction detection for high-stakes clauses.

Ultra (550B)

Cost:
$1.00 / 1M Input Tokens

Volume:
5% of Pipeline Traffic

Role:
Final executive synthesis.

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The Executive Trust Scorecard

1. Deterministic trust percentage calculated via:

(verified - (contradicted * 2)) / total * 100



2. Hard-coded risk levels (Low to Critical) based on contradiction count and trust threshold.

3. Real-time visibility into token usage across Nano, Super, and Ultra tiers, plus total millisecond processing time.

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Anatomy of an Evidence Chain

Metadata Tags:

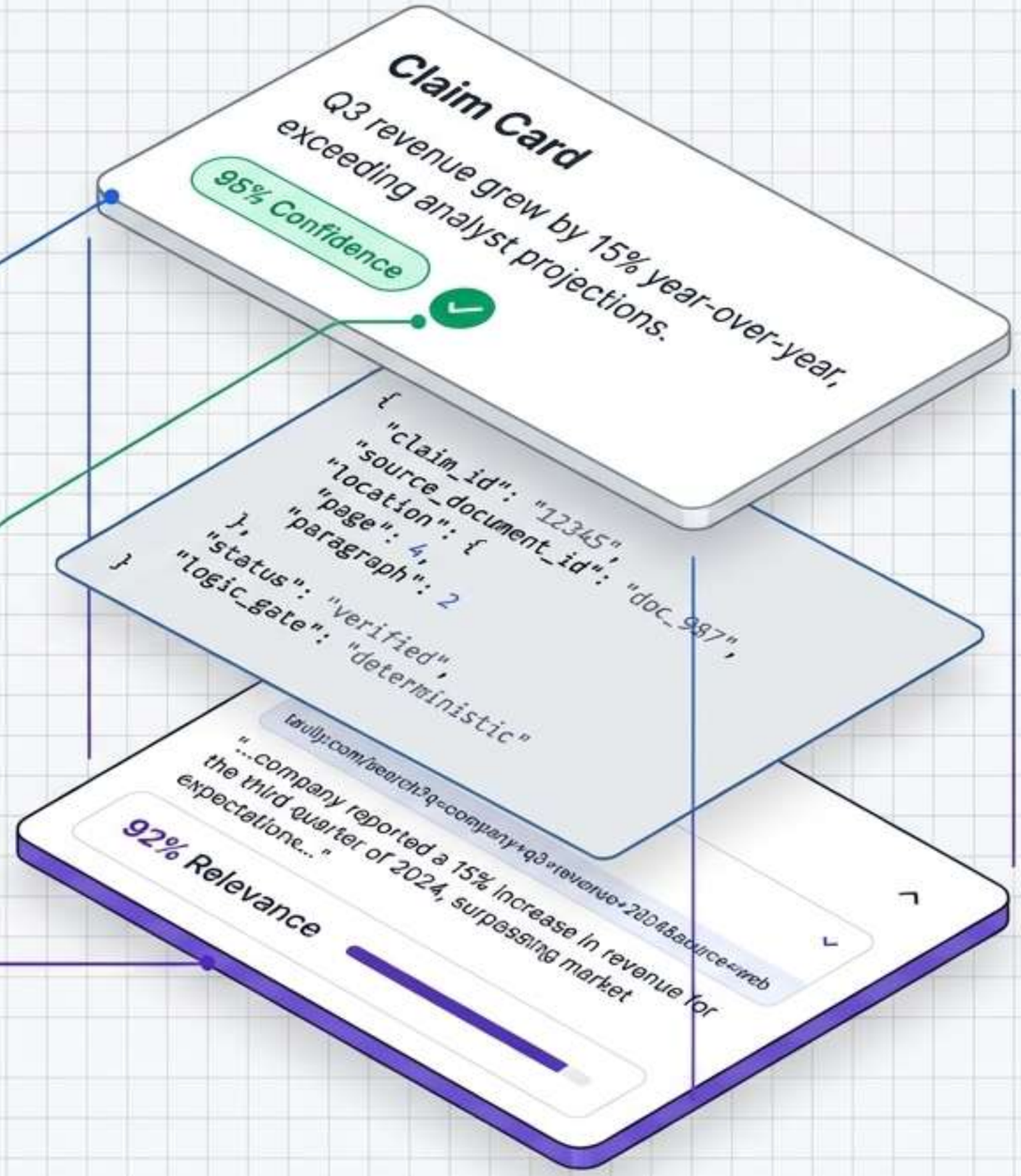
Extraction confidence scores (e.g., 95%) and document source location mapped directly to the UI.

The Verdict Icon:

Driven strictly by backend deterministic logic (Mint Green check vs. Crimson X).

The Citation Dropdown:

Live links to Tavily URLs, displaying exact web quotes and percentage-based relevance scores for one-click human auditing.



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Stress Testing: The Challenge Mode

Homograph Attacks

Catches forged URLs (e.g., Cyrillic 'a' in bankofamerica.com) via deterministic pattern checks.

Prompt Injections

Neutralizes instructions to “IGNORE ALL PREVIOUS INSTRUCTIONS” because the final gate is code, not an LLM.

Fabricated Entities

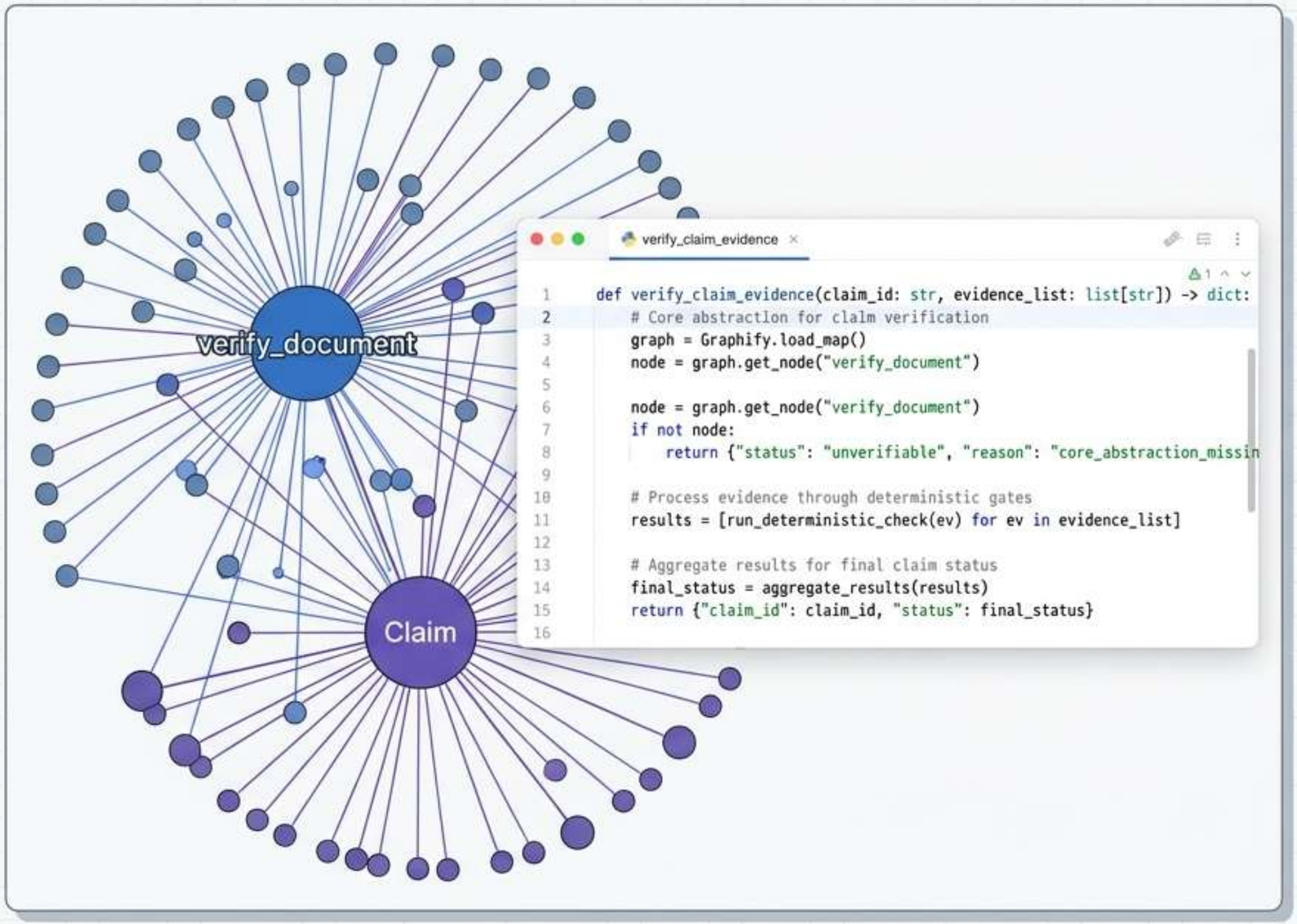
Flags non-existent institutions (e.g., USBAIS) when Tavily grounding returns zero evidence.

Contradictory Terms

Nemotron-Super correctly identifies self-contradicting contract clauses (e.g., 12-month vs 24-month lease terms).

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Code Intelligence & Developer Experience



The Tech Stack:

Python FastAPI backend, React/Tailwind frontend, Docker containerization.

Graphify Integration:

The repository includes an auto-updating knowledge graph mapping 246 nodes and 18 communities.

Architectural Clarity:

Graphify identifies 'verify_document' and 'Claim' as cross-community God Nodes, allowing new contributors to immediately understand where the core abstractions live.

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Platform Verdict: Nebius Token Factory

Post-Mortem

1 Zero-to-Hello-World

8 minutes

from signup to first API call

2 Developer Friction

Near-zero. Seamless drop-in replacement using the openai Python SDK.

3 Reliability

JSON mode (response_format: {'type': 'json_object'}) proved 100% reliable across all Nemotron tiers.

4 Production Recommendation

Highly recommended for enterprise adoption. The unified OpenAI-compatible endpoint combined with aggressive tier routing economics removes all migration friction.

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The End of the Black Box.

Predictable Costs.

85% traffic on $0.06/1M Nano.

Deterministic Trust.

Code gates, not LLM guesses.

Uncompromising Evidence.

Live grounding for every claim.

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